

4–5 Find the voltage gain v_O/v_S in Figure P4–5.

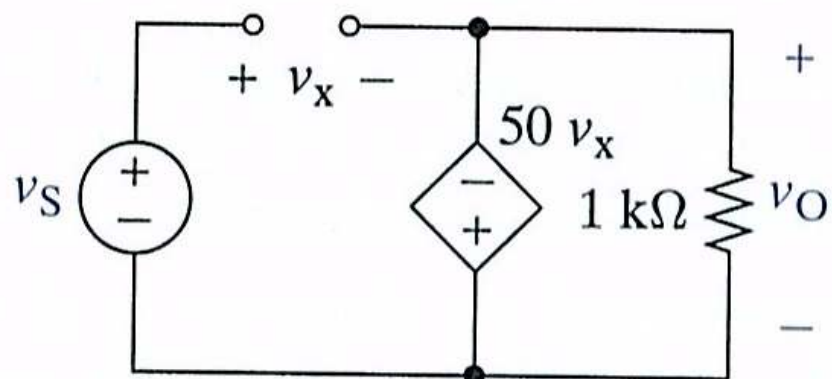


FIGURE P4–5

4–6 Find an expression for the current gain i_O/i_S in Figure P4–6.
Hint: Apply KCL at node A.

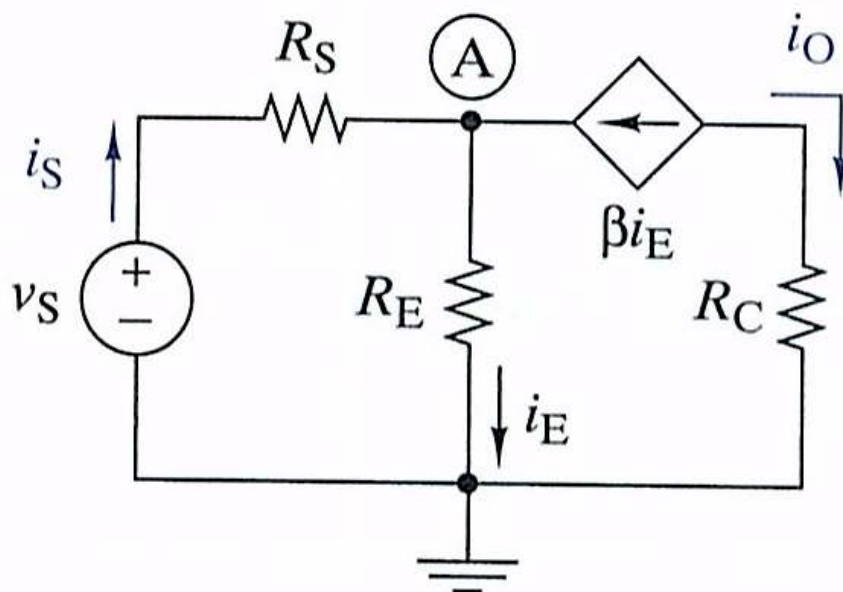


FIGURE P4–6

4-23 Find the voltage gain of each OP AMP circuit shown in Figure P4-23.

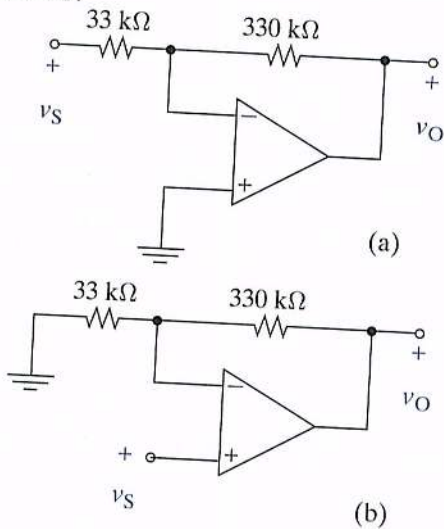


FIGURE P4-23

4-24 **D** Considering simplicity and standard 10% tolerance resistors as major constraints, design OP AMP circuits that produce the following voltage gains $\pm 10\%$: -100 , $+200$, $+1$, -0.5 , $+0.5$.

circuits to achieve this? Explain why.

4-27 (a) Find the voltage gain v_O/v_S in Figure P4-27.

(b) Validate your answer by simulating the circuit in OrCAD.

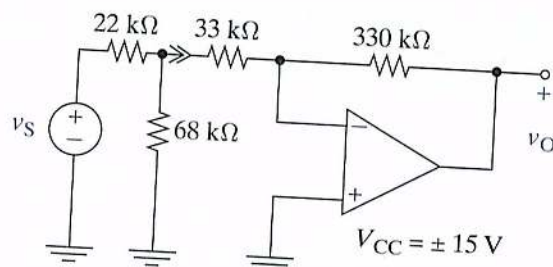


FIGURE P4-27

4-28 What is the range of the gain v_O/v_S in Figure P4-28?

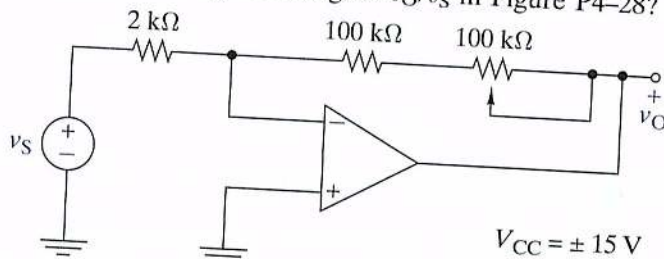


FIGURE P4-28

4-29 **D** Design a simple OP AMP circuit that has a variable gain from +10 to +100.

4-30 For the circuit in Figure P4-30:

- (a) Find v_O in terms v_S .
 (b) Find i_O for $v_S = 1$ V. Repeat for $v_S = 3$ V.

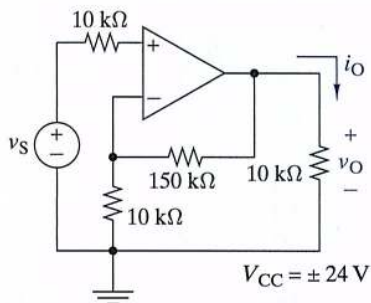


FIGURE P4-30

4-31 For the circuit in Figure P4-31:

- (a) Find v_O in terms v_S .
 (b) Find i_O for $v_S = 0.5$ V. Repeat for $v_S = 2$ V.

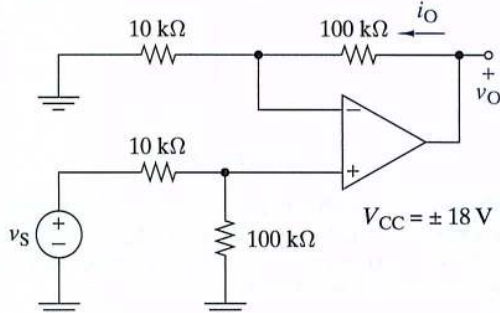


FIGURE P4-31

4-32 **E** A young designer needed to amplify a 2-V signal by the factors of 1, 5, and 10. Find the problem with the design shown in Figure P4-32. Recommend a fix.

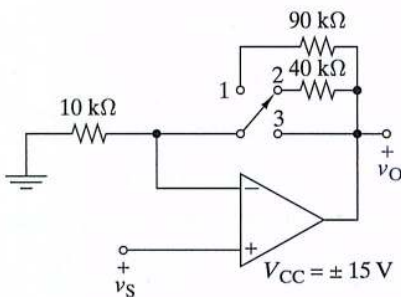


FIGURE P4-32

4-33 **D** Design two circuits to produce the following output: $v_O = 2v_1 - 3v_2$.

- (a) In your first design use a standard subtractor.
 (b) In your second design both inputs must be into high input resistance amplifiers to avoid loading.

4-34 For the circuit in Figure P4-34:

- (a) Find v_O in terms of the inputs v_1 and v_2 .
 (b) If $v_1 = 1$ V, what is the range of values v_2 can have without saturating the OP AMP?

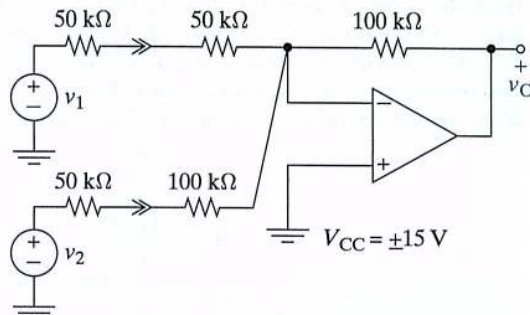


FIGURE P4-34

4-35 The input-output relationship for a three-input inverting summer is

$$v_O = -[v_1 + 10v_2 + 100v_3]$$

The resistance of the feedback resistor is 100 kΩ. Find the values of the input resistors R_1 , R_2 , and R_3 .

4-36 Find v_O in terms of the inputs v_1 and v_2 in Figure P4-36.

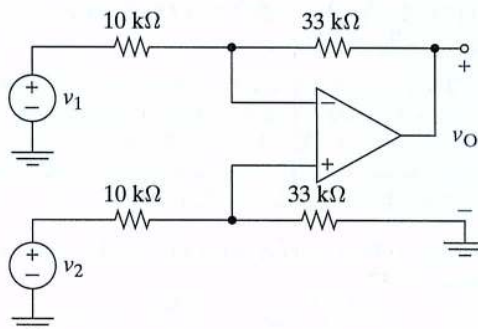


FIGURE P4-36

4-37 The switch in Figure P4-37 is open, find v_O in terms of the inputs v_{S1} and v_{S2} . Repeat with the switch closed.

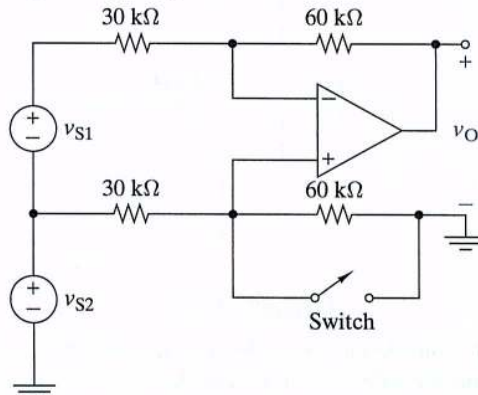


FIGURE P4-37

- 4-39 **E** It is claimed that $v_O = v_S$ when the switch is closed in Figure P4-39 and that $v_O = -v_S$ when the switch is open. Prove or disprove this claim.

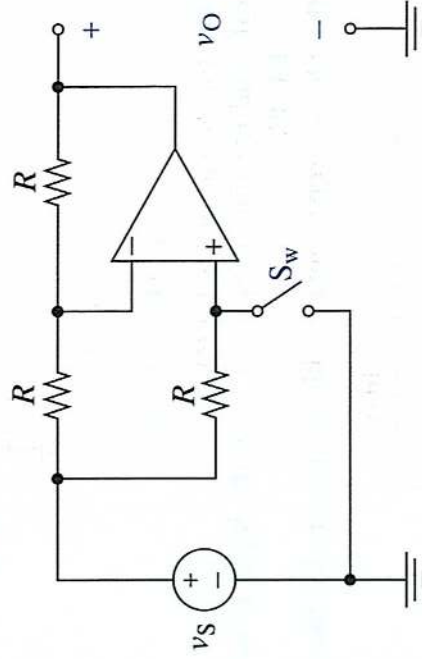


FIGURE P4-39

- 4-42 Use node voltage analysis in Figure P4-42 to show that $i_O = -v_S/2R$ regardless of the load. That is, show that the circuit is a voltage-controlled current source.

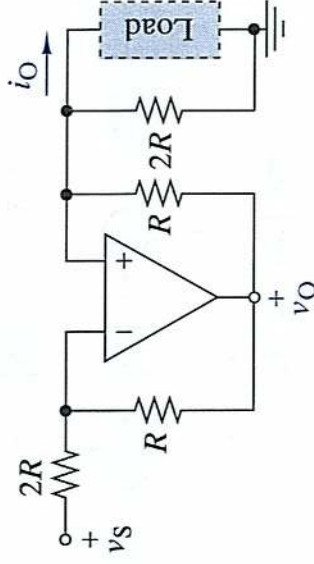


FIGURE P4-42

- 4-43 For the circuit of Figure P4-43:
(a) Find the output in terms of v_1 and v_2 .